

Nathen Carey

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Summary

Purdue University graduate with aeronautical and astronautical engineering. 2+ years internship experience and six months experience in CFD hypersonics research. Seeking an entry level engineering position. Dedicated engineer with the ability to communicate excellently with others. Eager to learn from professionals in engineering fields and utilize experience to succeed in future positions of employment. Avid musical enthusiast; proficient in 4 instruments.

Education

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| Purdue University , West Lafayette | Jan 2025 - May 2026 |
| <ul style="list-style-type: none">• Masters of Science in Aeronautics and Astronautics | |
| Purdue University , West Lafayette | Aug 2021 - Dec 2024 |
| <ul style="list-style-type: none">• Bachelor of Science in Aeronautical and Astronautical Engineering. | |

Previous Employment

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| Reditus Space Hypersonic CFD Research Assistant | Jan 2026 - May 2026 |
| <ul style="list-style-type: none">• Utilized ANSYS Fluent, SU2, Pointwise, and Techplot to assist hypersonic heat flux model analysis.• Gained practical research experience in workflows involving CFD, including meshing, simulation, and data analysis.• Operated on university Linux clusters to run RANS hypersonic simulations and analyze how different heat flux models perform compared to DNS datasets. | |
| Engine/Aircraft Performance Intern, JetZero | June 2024 - Present |
| <ul style="list-style-type: none">• Collaborated with engine companies including Pratt & Whitney and Rolls Royce to evaluate key engine performance parameters to best select propulsion systems for a Blended Wing Body aircraft.• Implemented several custom engine decks into an aircraft sizing code to evaluate aerodynamic performance of blended wing body aircraft with various engine selections.• Designed multiple blended wing body aircraft to best fit various GE engines. Ran performance on these designs to study effects of bypass ratio and lapse rates on engines fitted to BWBs.• Lead JetZero Propulsion and Aircraft Performance teams to create Rolls-Royce RB3142 Engine performance tables from the NPSS Model received from the NGAS Engine RFI. | |

Engineering Projects

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| Purdue Senior Design-Build-Fly | Fall 2024 - Winter 2024 |
| <ul style="list-style-type: none">• Performed trade studies to design and build a fixed wing remote controlled aircraft.• Wrote an extensive MATLAB aircraft sizing code to predict weights, drag, propulsion, and even simulate select stages of the DBF mission.• Utilized XFLR5 to perform aerodynamic optimization of an aircraft, and presented findings in professional settings to receive feedback from Purdue professors. | |
| High Speed Flow Simulation Project | Fall 2024 - Winter 2024 |
| <ul style="list-style-type: none">• Operated ANSYS Fluent to perform a grid refinement study on a supersonic curved wall geometry.• Analyzed supersonic flow and its interactions with oblique shocks, expansion fans, and a turbulent boundary layer.• Presented findings of study in a research paper style format for comparison to published results. | |

Skills

- Industry experience using NPSS customer decks and in creating engine models on AxStream.
- Extensive MATLAB experience in industry and research applications.
- Firm understanding of simulation software including ANSYS Fluent, XFLR5, OpenVSP, SU2, GMAT, and Free Flyer.